Part I: Communication Networks

```
s =

  struct with fields:

      degree: [5×1 double]
       edges: 4
    diameter: 2


s =

  struct with fields:

      degree: [5×1 double]
       edges: 5
    diameter: 2


s =

  struct with fields:

      degree: [5×1 double]
       edges: 10
    diameter: 1


s =

  struct with fields:

      degree: [6×1 double]
       edges: 6
    diameter: Inf


X =

     0     1     2     3
     1     1     1     1

     0

r_min = 1.2
T4 edges = 874, ring k = 6 (ring edges = 900)
                   Edges    MeanDeg    MaxDeg    Diam
                   _____    _______    ______    ____

    T1 Complete    11175       149       149       1
```

```
    T2 Star          149    1.9867      149        2
    T3 Ring          900        12       12       13
    T4 Disk          874    11.653       21       14

All W checks passed
               Random    Degree     Closest
               ______    _______    _______

    seed_1     17.693     10.705     9.2003
    seed_2     29.107     6.8411     5.3592
    seed_3     32.234     37.689     2.9157
    seed_4     43.652     68.795    0.34165
    seed_5     43.305     62.886    0.62493
    seed_6     18.055     21.094     6.2696
    seed_7     59.625      48.32     40.299
    seed_8     79.806     62.414     3.2273
    seed_9     58.418     6.4877     5.1796
    seed_10    49.163     27.344     16.999
    seed_11    45.042     44.991     19.628
    seed_12    83.097    0.91875    0.79427
    seed_13    73.961     31.215     3.9781
    seed_14    24.767     43.507     1.5416
    seed_15    62.438     64.643     5.7378
    seed_16    53.487     15.486     10.454
    seed_17    71.822     78.102     5.6529
    seed_18    21.078     37.109     10.898
    seed_19    60.814     69.075     3.9716
    seed_20    57.119     74.449     6.3433
    seed_21    54.369      85.16     1.8404
    seed_22    51.512      41.76     34.194
    seed_23    43.755     58.523    0.66638
    seed_24    75.116     78.575    0.18736
    seed_25    52.101     51.286     25.622
    seed_26    39.884     9.8202     8.6377
    seed_27     62.29      74.52      3.345
    seed_28    66.309     64.176     25.112
    seed_29    48.472     7.9235     2.1932
    seed_30    58.528     80.159     6.0174


Strategy             Mean   Median    Worst
Random              51.23    52.79    17.69
Highest degree      45.47    46.66     0.92
Closest to g         8.91     5.51     0.19
Random worst: seed 1, J = 17.69
Highest degree worst: seed 12, J = 0.92
Closest to g worst: seed 24, J = 0.19

ans =

   6.7927e-08


P0 =
```

```
    0.2652    0.2516
   -2.1646    1.8456
   -1.9007    0.6286
    2.0750   -0.4884
    0.3024   -0.0622


complete =

     0     1     1     1     1
     1     0     1     1     1
     1     1     0     1     1
     1     1     1     0     1
     1     1     1     1     0


star =

     0     1     1     1     1
     1     0     0     0     0
     1     0     0     0     0
     1     0     0     0     0
     1     0     0     0     0


ring =

     0     1     0     0     1
     1     0     1     0     0
     0     1     0     1     0
     0     0     1     0     1
     1     0     0     1     0

r_min = 2.2

disk =

     0     0     1     1     1
     0     0     1     0     0
     1     1     0     0     0
     1     0     0     0     1
     1     0     0     1     0


disk_big =

     0     0     1     1     1
     0     0     1     0     0
     1     1     0     0     1
     1     0     0     0     1
     1     0     1     1     0

Network 1:
```

```
Weight Matrix for Network A:
    0.2000    0.2000    0.2000    0.2000    0.2000
    0.2000    0.2000    0.2000    0.2000    0.2000
    0.2000    0.2000    0.2000    0.2000    0.2000
    0.2000    0.2000    0.2000    0.2000    0.2000
    0.2000    0.2000    0.2000    0.2000    0.2000

Weight Matrix for Network B:
         0    0.2500    0.2500    0.2500    0.2500
    0.2500         0    0.2500    0.2500    0.2500
    0.2500    0.2500         0    0.2500    0.2500
    0.2500    0.2500    0.2500         0    0.2500
    0.2500    0.2500    0.2500    0.2500         0

Network 2:
Weight Matrix for Network A:
    0.2000    0.2000    0.2000    0.2000    0.2000
    0.5000    0.5000         0         0         0
    0.5000         0    0.5000         0         0
    0.5000         0         0    0.5000         0
    0.5000         0         0         0    0.5000

Weight Matrix for Network B:
         0    0.2500    0.2500    0.2500    0.2500
    0.2500    0.7500         0         0         0
    0.2500         0    0.7500         0         0
    0.2500         0         0    0.7500         0
    0.2500         0         0         0    0.7500

Network 3:
Weight Matrix for Network A:
    0.3333    0.3333         0         0    0.3333
    0.3333    0.3333    0.3333         0         0
         0    0.3333    0.3333    0.3333         0
         0         0    0.3333    0.3333    0.3333
    0.3333         0         0    0.3333    0.3333

Weight Matrix for Network B:
         0    0.5000         0         0    0.5000
    0.5000         0    0.5000         0         0
         0    0.5000         0    0.5000         0
         0         0    0.5000         0    0.5000
    0.5000         0         0    0.5000         0

Network 4:
Weight Matrix for Network A:
    0.2500         0    0.2500    0.2500    0.2500
         0    0.5000    0.5000         0         0
    0.3333    0.3333    0.3333         0         0
    0.3333         0         0    0.3333    0.3333
    0.3333         0         0    0.3333    0.3333

Weight Matrix for Network B:
         0         0    0.3333    0.3333    0.3333
```

```
         0    0.6667    0.3333         0         0
    0.3333    0.3333    0.3333         0         0
    0.3333         0         0    0.3333    0.3333
    0.3333         0         0    0.3333    0.3333

Network 5:
Weight Matrix for Network A:
    0.2500         0    0.2500    0.2500    0.2500
         0    0.5000    0.5000         0         0
    0.2500    0.2500    0.2500         0    0.2500
    0.3333         0         0    0.3333    0.3333
    0.2500         0    0.2500    0.2500    0.2500

Weight Matrix for Network B:
         0         0    0.3333    0.3333    0.3333
         0    0.6667    0.3333         0         0
    0.3333    0.3333         0         0    0.3333
    0.3333         0         0    0.3333    0.3333
    0.3333         0    0.3333    0.3333         0

graph_stats, Network 1:

st =

  struct with fields:

      degree: [5×1 double]
       edges: 10
    diameter: 1

graph_stats, Network 2:

st =

  struct with fields:

      degree: [5×1 double]
       edges: 4
    diameter: 2

graph_stats, Network 3:

st =

  struct with fields:

      degree: [5×1 double]
       edges: 5
    diameter: 2

graph_stats, Network 4:

st =
```

```
  struct with fields:

      degree: [5×1 double]
       edges: 5
    diameter: 3

graph_stats, Network 5:

st =

  struct with fields:

      degree: [5×1 double]
       edges: 6
    diameter: 3


st_disc =

  struct with fields:

      degree: [6×1 double]
       edges: 6
    diameter: Inf

Row sums, Network 1 (WA, WB):
     1     1
     1     1
     1     1
     1     1
     1     1

Row sums, Network 2 (WA, WB):
     1     1
     1     1
     1     1
     1     1
     1     1

Row sums, Network 3 (WA, WB):
     1     1
     1     1
     1     1
     1     1
     1     1

Row sums, Network 4 (WA, WB):
     1     1
     1     1
     1     1
     1     1
     1     1

Row sums, Network 5 (WA, WB):
```

```
     1     1
     1     1
     1     1
     1     1
     1     1

Network 1, Protocol A:
    0.2652   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845
   -2.1646   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845
   -1.9007   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845
    2.0750   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845
    0.3024   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845   -0.2845

Network 1, Protocol B:
    0.2652   -0.4220   -0.2502   -0.2931   -0.2824   -0.2851   -0.2845
   -2.1646    0.1855   -0.4020   -0.2552   -0.2919   -0.2827   -0.2845
   -1.9007    0.1195   -0.3855   -0.2593   -0.2908   -0.2830   -0.2845
    2.0750   -0.8744   -0.1371   -0.3214   -0.2753   -0.2868   -0.2845
    0.3024   -0.4313   -0.2479   -0.2937   -0.2822   -0.2851   -0.2845

Network 2, Protocol A:
    0.2652   -0.2845   -0.1196   -0.1691   -0.1543   -0.1587   -0.1577
   -2.1646   -0.9497   -0.6171   -0.3684   -0.2687   -0.2115   -0.1577
   -1.9007   -0.8178   -0.5512   -0.3354   -0.2522   -0.2032   -0.1577
    2.0750    1.1701    0.4428    0.1616   -0.0038   -0.0790   -0.1577
    0.3024    0.2838   -0.0004   -0.0600   -0.1145   -0.1344   -0.1577

Network 2, Protocol B:
    0.2652   -0.4220   -0.2502   -0.2931   -0.2824   -0.2851   -0.2845
   -2.1646   -1.5571   -1.2733   -1.0175   -0.8364   -0.6979   -0.2848
   -1.9007   -1.3592   -1.1249   -0.9062   -0.7530   -0.6353   -0.2848
    2.0750    1.6226    1.1114    0.7710    0.5050    0.3081   -0.2841
    0.3024    0.2931    0.1143    0.0232   -0.0559   -0.1125   -0.2844

Network 3, Protocol A:
    0.2652   -0.5323   -0.3061   -0.3192   -0.2985   -0.2930   -0.2845
   -2.1646   -1.2667   -0.8208   -0.5724   -0.4401   -0.3684   -0.2845
   -1.9007   -0.6634   -0.5904   -0.4286   -0.3665   -0.3279   -0.2845
    2.0750    0.1589    0.1254   -0.0986   -0.1770   -0.2280   -0.2845
    0.3024    0.8809    0.1691   -0.0038   -0.1406   -0.2054   -0.2845

Network 3, Protocol B:
    0.2652   -0.9311    0.1762   -0.6765    0.0266   -0.5381   -0.2833
   -2.1646   -0.8178   -0.4879   -0.3161   -0.3196   -0.2749   -0.2846
   -1.9007   -0.0448   -0.8085    0.0374   -0.5765   -0.0581   -0.2857
    2.0750   -0.7992    0.5627   -0.8368    0.2034   -0.6666   -0.2826
    0.3024    1.1701   -0.8651    0.3694   -0.7567    0.1150   -0.2865

Network 4, Protocol A:
    0.2652    0.1855    0.1701    0.1076    0.0618    0.0236   -0.1221
   -2.1646   -2.0326   -1.6497   -1.3438   -1.0915   -0.8916   -0.1249
   -1.9007   -1.2667   -1.0380   -0.8392   -0.6918   -0.5738   -0.1239
    2.0750    0.8809    0.6491    0.4894    0.3621    0.2620   -0.1214
    0.3024    0.8809    0.6491    0.4894    0.3621    0.2620   -0.1214
```

```
Network 4, Protocol B:
    0.2652    0.1589    0.1650    0.0730    0.0209   -0.0348   -0.2824
   -2.1646   -2.0766   -1.8066   -1.5583   -1.3392   -1.1579   -0.2921
   -1.9007   -1.2667   -1.0615   -0.9010   -0.7954   -0.7046   -0.2882
    2.0750    0.8809    0.6402    0.4818    0.3455    0.2373   -0.2800
    0.3024    0.8809    0.6402    0.4818    0.3455    0.2373   -0.2800

Network 5, Protocol A:
    0.2652    0.1855    0.0943   -0.0070   -0.0717   -0.1160   -0.2021
   -2.1646   -2.0326   -1.4535   -1.0438   -0.7592   -0.5712   -0.2022
   -1.9007   -0.8744   -0.6340   -0.4747   -0.3831   -0.3214   -0.2022
    2.0750    0.8809    0.4173    0.2020    0.0627   -0.0269   -0.2021
    0.3024    0.1855    0.0943   -0.0070   -0.0717   -0.1160   -0.2021

Network 5, Protocol B:
    0.2652    0.1589    0.1650   -0.0086   -0.0587   -0.1341   -0.2845
   -2.1646   -2.0766   -1.5619   -1.2380   -0.9618   -0.7808   -0.2847
   -1.9007   -0.5323   -0.5904   -0.4092   -0.4189   -0.3596   -0.2846
    2.0750    0.8809    0.3954    0.2432    0.0749   -0.0140   -0.2845
    0.3024    0.1465    0.1691   -0.0100   -0.0582   -0.1342   -0.2845


ans =

   -0.2845


ans =

   -0.1226

Spreads at each time step should decrease:
    3.9397

    2.5329

    1.6789

    1.1138

    0.7370

    0.4878

Spreads at final time step should be near zero:
   3.2690e-08

   2.1630e-08

   1.4313e-08

   9.4705e-09
```

```
   6.2666e-09
```

```
g_test =

    5.7736    5.3920
```

```
Xa =

    0.2652    0.1855    0.6232    0.9515    1.1977    1.3824    1.5209
   -2.1646   -2.0326   -1.9667   -1.9337   -1.9172   -1.9090   -1.9048
   -1.9007   -1.9007   -1.9007   -1.9007   -1.9007   -1.9007   -1.9007
    2.0750    0.8809    1.4645    1.9023    2.2306    2.4768    2.6615
    0.3024    3.3272    3.6191    3.8379    4.0021    4.1252    4.2175
```

```
Ya =

    0.2516    0.0824    0.8144    1.3634    1.7751    2.0839    2.3155
    1.8456    1.2371    0.9329    0.7808    0.7047    0.6667    0.6477
    0.6286    0.6286    0.6286    0.6286    0.6286    0.6286    0.6286
   -0.4884   -0.0997    0.8763    1.6083    2.1572    2.5690    2.8778
   -0.0622    2.6461    3.1341    3.5001    3.7746    3.9805    4.1349
```

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